Protocol Intelligence Report

Classification

Internal review draft

File: trial_protocol_report.pdf

Structured benchmarking, decision-grade analysis, PubMed evidence enrichment, and audit-ready reporting for clinical trial design review.

Generated (UTC): 2026-03-19T17:56:59.964832Z

Prepared for clinical development, trial design, and operational planning review.

1. Executive Summary

This report provides a structured benchmarking assessment of the reviewed Phase 3b Interventional protocol targeting Asthma. The analysis draws on a matched cohort of **972 precedent studies** retrieved from the ClinicalTrials.gov registry, of which **626 reached completion** and **70 were disrupted** (terminated, suspended, or withdrawn). Evidence strength for this cohort is rated **Strong**.

Across the seven design domains assessed, the reviewed draft shows **a mixed precedent posture**. The completed-precedent fit score is **39.4%** compared to a disrupted-precedent fit of **41.4%**, yielding a net precedent gap of **-2.0%**. The overall posture classification is: **Mixed precedent posture**.

All outputs are intended to support expert review and should be interpreted alongside clinical, statistical, operational, and regulatory judgment. No finding in this report constitutes a regulatory opinion or approval recommendation.

Field	Value
Matched cohort	972 studies
Completed comparators	626
Disrupted comparators	70
Evidence strength	Strong
Completed design fit	39.4%
Disrupted design fit	41.4%
Net precedent gap	-2.0%
Overall posture	Mixed precedent posture

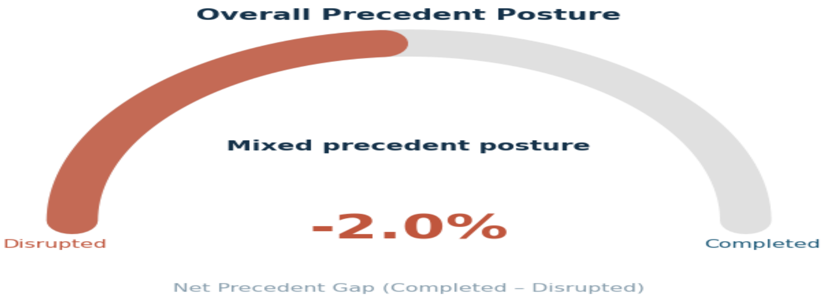

Figure 1. Overall Precedent Posture Gauge — net gap between completed and disrupted design fit.

2. Senior Decision Signals

The following table summarises the five critical decision areas assessed in this benchmark. Each signal should be reviewed by the senior design team before moving to protocol finalisation or governance submission.

Decision Area	Signal	Implication	Evidence
Evidence base	Strong	Confidence is highest when completed and disrupted comparators are both adequately represented.	972 total studies | 626 completed | 70 disrupted.
Overall posture	Mixed precedent posture	This is the clearest top-line read on whether the draft is behaving more like successful or disrupted precedent.	Completed fit 39.4% | Disrupted fit 41.4%.
Enrollment plan	Insufficient evidence	Enrollment targets outside the completed range need stronger feasibility justification.	Target n/a | Completed IQR 40.0 to 350.5.
Execution exposure	Moderate disruption exposure	Higher disruption prevalence should trigger more explicit mitigation planning.	Disrupted-status share 7.2%.
Protocol completeness	Complete	Missing design fields make the benchmark less trustworthy and reduce report defensibility.	No missing core fields detected.
Lead action	Monitor	No strong design outlier was detected in the current benchmark snapshot.	Continue with clinical, statistical, operational, and regulatory review before final sign-off.

3. Executive Action Register

Actions are prioritised as **High** (requires resolution before sign-off), **Medium** (should be addressed before governance), **Monitor** (watch as evidence evolves), or **Preserve** (design choice is well-supported — maintain it). This run generated **1 action(s)**.

Priority	Category	Action	Recommendation	Evidence
Monitor	Overall review	Monitor	No strong design outlier was detected in the current benchmark snapshot.	Continue with clinical, statistical, operational, and regulatory review before final sign-off.

4. Reviewed Protocol Profile

The following fields were extracted from the uploaded protocol document and reviewed by the study team. Extraction confidence: **high**. Profile status: **Reviewed**.

Protocol Title: A Randomized, Double -blind, Single Dose, 2 -way Crossover Study to Assess Bronchospasm Potentially Induced by the Hydrofluoroolefin (HFO) Metered Dose Inhaler (MDI) Propellant as Compared with the Hydrofluoroalkane (HFA) MDI Propellant in Participants with Asthma, Well Controlled or Partially Controlled on Short -acting beta 2-agonists (SABA) with or without Low -Dose Inhaled Corticosteroids (ICS)

Field	Value
Condition	Asthma
Sponsor	AstraZeneca AB
Study Type	Interventional
Phase	Phase 3b
Planned Enrollment	Not provided
Arms Count	Not provided
Allocation	Randomized
Masking	Double
Intervention Model	Crossover Assignment
Primary Purpose	Treatment
Geography Focus	Not provided
Start Date	08 December 2022
Completion Date	Not provided
Endpoint Focus	Other

Primary Endpoints

change from baseline in FEV 1 AUC 0-15 min post-dose

Secondary Endpoints

cumulative incidence of bronchospasm events
safety and tolerability of HFO MDI as compared to HFA MDI in participants with asthma, evaluated in terms of AEs

Target Population & Eligibility Criteria

Participants are eligible to be included in the study only if all of the following criteria apply:
1 Male and female participant must be 18 to 45 years of age inclusive, at the time of signing the informed consent form (ICF).
2 Participants who have a documented history of physician-diagnosed asthma ≥12 months prior to Visit 1, according to GINA guidelines (GINA 2022).
3 Participants who are well controlled or partially controlled on their current treatment for asthma, including, low-dose ICS daily or low-dose ICS/formoterol as needed (not approved in the US), or SABA as needed, or low-dose ICS whenever SABA as needed is used (low-dose ICS as defined by GINA 2022 inTable 4), for 4 weeks prior to screening.
4 ACQ-5 total score < 1.5 at

5 A pre-bronchodilator FEV 1> 60% predicted normal value at
6 Demonstrate acceptable MDI administration technique.
7 Females must be not of childbearing potential, or should be using a form of highly effective birth control as defined below:
! Female participants: Women not of childbearing potential are defined as women who are either permanently sterilised (hysterectomy, bilateral oophorectomy, or bilateral salpingectomy), or who are postm enopausal. Women included in this study will be considered postmenopausal if they have been amenorrhoeic for 12 months or more following cessation of exogenous hormonal treatment and follicle-stimulating hormone levels in the postmenopausal range.
! Female participants of childbearing potential must use one highly effective form of birth control. A highly effective method of contraception is defined as one that can achieve a failure rate of less than 1% per year when used consistently and correctly. At enrolment, women of childbearing potential who are sexually active with a non-sterilised male partner should be stable on their chosen method of highly effective birth control, as defined below, and willing to remain on the birth control until at least 14 days after last dose of study intervention. Cessation of contraception after this point should be discussed with a responsible physician.
Periodic abstinence (calendar, sy mptothermal, post-ovulation methods),
withdrawal (coitus interruptus), spermicides only, and lactational amenorrhoea method are not acceptable methods of contraception. Female condom and male condom should not be used together. All women of childbearing potential must
have a negative serum pregnancy test result
! Highly effective birth control methods are listed below:
oTotal sexual abstinence is an acceptable method provided it is the usual
lifestyle of the participant (defined as refraining from heterosexual intercourse during the entire period of risk associated with the study treatments). Periodic abstinence (e .g., calendar, ovulation, symptothermal, post-ovulation methods), declaration of abstinence for the d

Intervention Description

HFO propellant only MDI; 4 inhalations per dose – test formulation

Comparator / Control Arm

HFA MDI delivered in participants with well controlled or partially controlled asthma defined as an ACQ-5 score < 1.5.

5. Precedent Posture Analysis

Overall precedent posture: **Mixed precedent posture** — based on 972 matched studies. The draft sits in a mixed precedent zone (completed fit 39.4% vs disrupted fit 41.4%). There is no strong directional signal, which means execution quality, operational planning, and individual domain choices will carry more decision weight than the aggregate posture.

6. Design Domain Alignment

No domains showed extreme divergence between completed and disrupted precedent. The design alignment is broadly balanced, though individual domain values should still be reviewed against the specific design rationale.

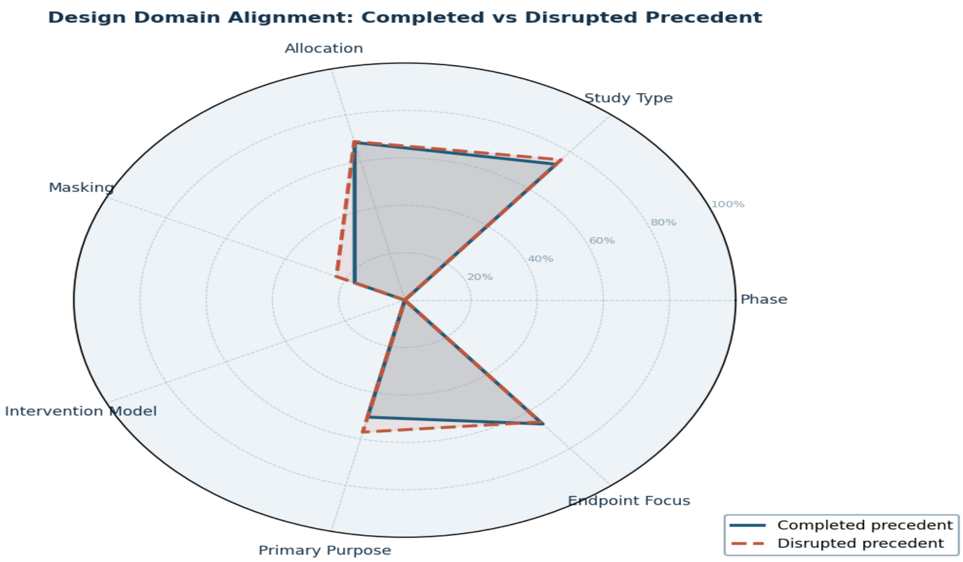

Figure 2. Design Domain Alignment Radar — completed precedent (teal filled) vs disrupted precedent (red dashed). Larger filled area = stronger alignment with successful trials.

Design Alignment Risk Signal Summary

Domain	Completed %	Disrupted %	Gap (C-D)	Signal
Phase	0.0%	0.0%	+0.0%	Mixed
Study Type	73.3%	75.7%	-2.4%	Mixed
Allocation	68.2%	68.6%	-0.4%	Mixed
Masking	16.8%	22.9%	-6.1%	Mixed
Intervention Model	0.0%	0.0%	+0.0%	Mixed
Primary Purpose	50.6%	57.1%	-6.5%	Mixed
Endpoint Focus	66.8%	65.7%	+1.1%	Mixed

Figure 3. Domain Alignment Risk Signal Summary — green = closer to completed precedent; red = closer to disrupted.

7. Design Differential Matrix

This matrix shows, domain by domain, the match rate the protocol has with completed and disrupted precedent, the net gap between them, and the signal classification. The 'Why It Matters' context helps reviewers prioritise which gaps need explicit narrative justification.

Domain	Protocol Choice	Completed %	Disrupted %	Gap %	Signal
Phase	Phase 3b	0.0	0.0	0.0	Weak precedent signal
Study Type	Interventional	73.3	75.7	-2.4	Mixed precedent
Allocation	Randomized	68.2	68.6	-0.4	Mixed precedent
Masking	Double	16.8	22.9	-6.1	Mixed precedent
Intervention Model	Crossover Assignment	0.0	0.0	0.0	Weak precedent signal
Primary Purpose	Treatment	50.6	57.1	-6.5	Mixed precedent
Endpoint Focus	Other	66.8	65.7	1.1	Mixed precedent

8. Enrollment Benchmark

Enrollment benchmarking data is insufficient. The protocol's planned enrollment may not have been extracted, or the comparator cohort is too thin.

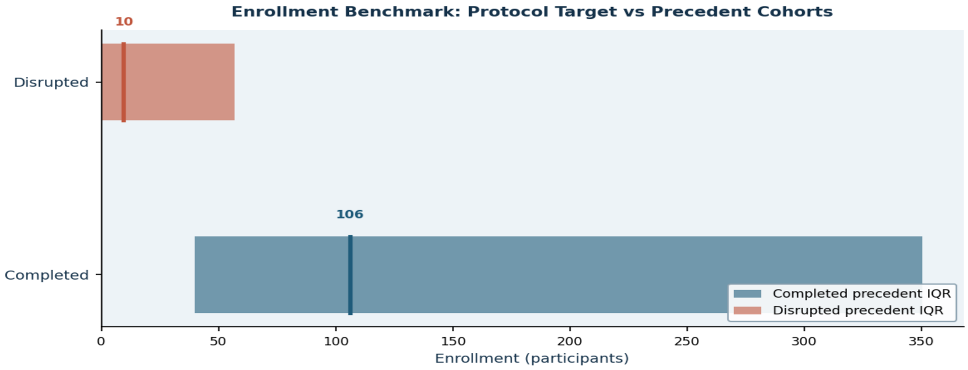

Figure 4. Enrollment Benchmark — IQR boxes show 25th–75th percentile range; centre line is the median; amber dashed = protocol target.

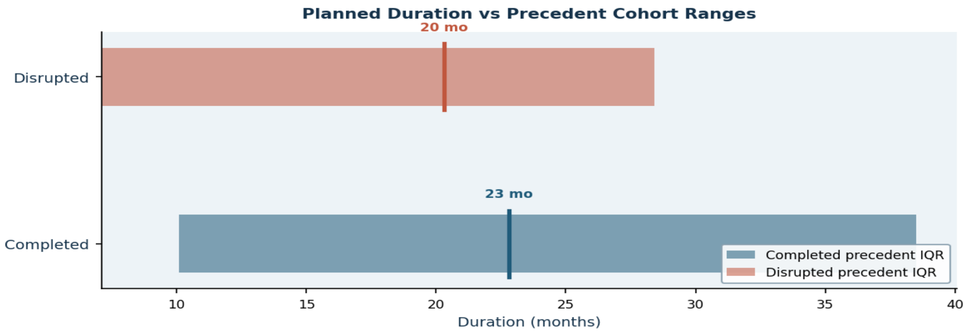

Figure 5. Planned Duration vs Precedent Cohort Ranges — amber dashed line = protocol planned duration.

9. Comparator Cohort Definition

The following table describes how the comparator cohort was constructed. Only studies matching the clinical condition specified in the protocol were included. Status classification (completed vs disrupted) follows ClinicalTrials.gov status fields.

Metric	Value
Clinical condition	Asthma
Matched trial count	972
Completed comparators	626
Disrupted comparators	70
Active comparators	143
Evidence strength	Strong
Risk-status share	7.2%
Median site count	1.0
Median country count	1.0

10. Success vs Disruption Benchmark

This table presents, for each design parameter, the protocol's value alongside the comparable range from completed and disrupted trials. The signal column classifies the protocol's position relative to completed-trial precedent.

Domain	Protocol Draft	Completed Precedent	Disrupted Precedent	Signal
Enrollment	Not provided	Median 106.0 | IQR 40.0 to 350.5	Median 9.5 | IQR 0.0 to 57.0	Insufficient evidence
Planned Duration	Not provided	Median 22.8 mo | IQR 10.1 to 38.5 mo	Median 20.3 mo | IQR 7.1 to 28.4 mo	Insufficient evidence
Phase	Phase 3b	0.0%	0.0%	Weak precedent signal
Study Type	Interventional	73.3%	75.7%	Mixed precedent
Allocation	Randomized	68.2%	68.6%	Mixed precedent
Masking	Double	16.8%	22.9%	Mixed precedent
Intervention Model	Crossover Assignment	0.0%	0.0%	Weak precedent signal
Primary Purpose	Treatment	50.6%	57.1%	Mixed precedent
Endpoint Focus	Other	66.8%	65.7%	Mixed precedent

11. Decision Scorecard

Quantitative scorecard of all computed metrics. These numbers underpin the signals and recommendations in this report. Any metric showing 'Not available' reflects a data gap in either the protocol or the comparator cohort.

Metric	Value
Matched cohort size	972
Completed comparator count	626
Disrupted comparator count	70
Evidence strength	Strong
Completed precedent fit (%)	39.4
Disrupted precedent fit (%)	41.4
Net precedent gap (%)	-2.0
Overall posture	Mixed precedent posture
Completed enrollment IQR	40.0 to 350.5
Disrupted enrollment IQR	0.0 to 57.0
Completed median duration (months)	22.8
Disrupted median duration (months)	20.3
Median site count	1.0
Median country count	1.0

Metric	Value
Risk-status share (%)	7.2
Completed share (%)	64.4

12. Endpoint Category Evidence

Endpoint category distribution reveals what types of evidence completed and disrupted trials have relied upon in this condition. The protocol's current endpoint focus is highlighted. Aligning with completed-trial endpoint patterns is associated with stronger design credibility.

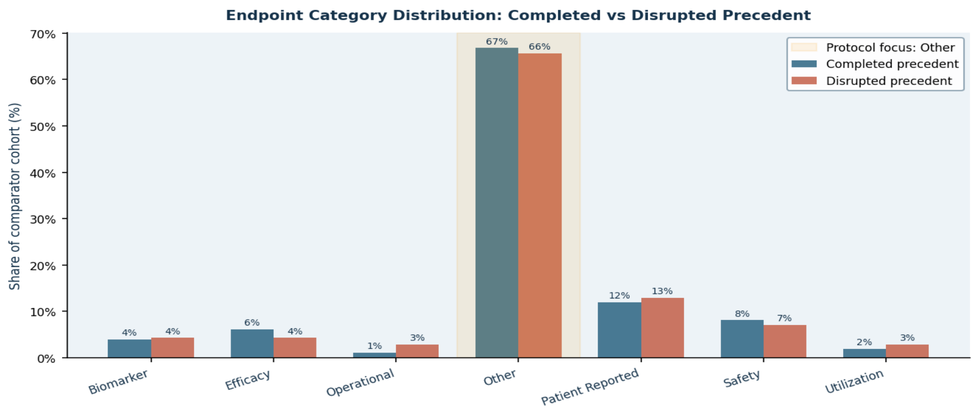

Figure 6. Endpoint Category Distribution — completed precedent (teal) vs disrupted precedent (red). Amber shading = protocol endpoint focus.

Endpoint Category	Completed Share (%)	Disrupted Share (%)	Net Gap (%)
Other	66.8	65.7	1.1
Patient Reported	12.0	12.9	-0.9
Safety	8.1	7.1	1.0
Efficacy	6.1	4.3	1.8
Biomarker	4.0	4.3	-0.3
Utilization	1.9	2.9	-1.0

13. PubMed Literature Evidence

8 peer-reviewed article(s) were retrieved from PubMed/MEDLINE via the NCBI E-utilities free API to provide literature context for this trial design. All citations below are fully traceable — each entry includes the PubMed ID (PMID), authors, journal, year of publication, and the query used to retrieve it. The abstract excerpts are provided to help the reviewer judge relevance quickly. Full articles can be accessed at pubmed.ncbi.nlm.nih.gov/[PMID].

PMID	Authors	Year	Journal	Title	Endpoint Focus
39465893	Rayner DG et al.	2025	JAMA	Inhaled Reliever Therapies for Asthma: A Systematic Review and Meta-Analysi	General
36538979	Nopsopon T et al.	2023	The Journal of allergy and clinical	Comparative efficacy of tezepelumab to mepolizumab, benralizumab, and dupil	Efficacy
32350100	Hansen ESH et al.	2020	The European respiratory journal	Effect of aerobic exercise training on asthma in adults: a systematic revie	General
39421966	Fedora K et al.	2024	Annals of medicine	Vitamin D supplementation decrease asthma exacerbations in children: a syst	General

PMID	Authors	Year	Journal	Title	Endpoint Focus
37286218	Makrufardi F et al.	2023	European respiratory review : an of	Extreme weather and asthma: a systematic review and meta-analysis.	Efficacy
38657997	Kyriakopoulos C et al.	2024	European respiratory review : an of	Biologic agents licensed for severe asthma: a systematic review and meta-an	Efficacy
39384302	Manti S et al.	2024	European respiratory review : an of	Epidemiology of severe asthma in children: a systematic review and meta-ana	General
39549709	Shackleford A et al.	2025	The Lancet. Respiratory medicine	Clinical remission attainment, definitions, and correlates among patients w	Efficacy

Abstract Excerpts

Inhaled Reliever Therapies for Asthma: A Systematic Review and Meta-Analysis. — Rayner DG, Ferri DM, Guyatt GH | JAMA (2025) PMID: 39465893 | Query: "Asthma"[MeSH Terms] OR "Asthma"[Title/Abstract] AND ("clinical trial"[Publication Type] OR "randomized controlled trial"[Publication Type] OR "systematic review"[Publication Type] OR "meta-analysis"[Publication Type])

The optimal inhaled reliever therapy for asthma remains unclear. To compare short-acting β agonists (SABA) alone with SABA combined with inhaled corticosteroids (ICS) and with the fast-onset, long-acting β agonist formoterol combined with ICS for asthma. The MEDLINE, Embase, and CENTRAL databases were searched from January 1, 2020, to September 27, 2024, without language restrictions. Pairs of reviewers independently selected randomized clinical trials evaluating (1) SABA alone, (2) ICS with formoterol, and (3) ICS with SABA (combined or separate inhalers). Two reviewers independently extracte...

Comparative efficacy of tezepelumab to mepolizumab, benralizumab, and dupilumab in eosinophilic asthma: A Bayesian network meta-analysis. — Nopsopon T, Lassiter G, Chen ML | The Journal of allergy and clinical immunology (2023) PMID: 36538979 | Query: "Asthma"[MeSH Terms] OR "Asthma"[Title/Abstract] AND ("clinical trial"[Publication Type] OR "randomized controlled trial"[Publication Type] OR "systematic review"[Publication Type] OR "meta-analysis"[Publication Type])

It is unclear how the efficacy of tezepelumab, approved for the treatment of type 2 high and low asthma, compares to the efficacy of other biologics for type 2-high asthma. We sought to conduct an indirect comparison of tezepelumab to dupilumab, benralizumab, and mepolizumab in the treatment of eosinophilic asthma. The investigators conducted a systematic review and Bayesian network meta-analyses. They identified randomized controlled trials indexed in PubMed, Embase, or Cochrane Central Register of Controlled Trials (CENTRAL) between January 1, 2000, and August 12, 2022. Outcomes included exa...

Effect of aerobic exercise training on asthma in adults: a systematic review and meta-analysis. — Hansen ESH, Pitzner-Fabricius A, Toennesen LL | The European respiratory journal (2020) PMID: 32350100 | Query: "Asthma"[MeSH Terms] OR "Asthma"[Title/Abstract] AND ("clinical trial"[Publication Type] OR "randomized controlled trial"[Publication Type] OR "systematic review"[Publication Type] OR "meta-analysis"[Publication Type])

To evaluate the effect of aerobic exercise training on asthma control, lung function and airway inflammation in adults with asthma. Systematic review and meta-analysis. Randomised controlled trials investigating the effect of ≥8 weeks of aerobic exercise training on outcomes for asthma control, lung function and airway inflammation in adults with asthma were eligible for study. MEDLINE, Embase, CINAHL, PEDro and the Cochrane Central Register of Controlled Trials (CENTRAL) were searched up to April 3, 2019. Risk of bias was assessed using the Cochrane Risk of Bias Tool. We included 11 studies w...

Vitamin D supplementation decrease asthma exacerbations in children: a systematic review and meta-analysis of randomized controlled trials. — Fedora K, Setyoningrum RA, Aina Q | Annals of medicine (2024) PMID: 39421966 | Query: "Asthma"[MeSH Terms] OR "Asthma"[Title/Abstract] AND ("clinical trial"[Publication Type] OR "randomized controlled trial"[Publication Type] OR "systematic review"[Publication Type] OR "meta-analysis"[Publication Type])

Observational studies have linked low vitamin D (VD) levels to increased asthma attacks in children. Subsequent meta-analyses of adults and children revealed that VD treatment might benefit asthmatic patients by reducing the incidence of exacerbations. Therefore, this review aims to analyze the effects of VD supplementation in reducing asthma exacerbations in children. Published reports from PubMed, Cochrane, and Google Scholar were systematically searched until April 2023. The study protocol was registered in the PROSPERO database CRD42023411796. Randomized controlled trial studies were inclu...

14. Comparative Narrative

Matched 972 studies for Asthma, including 626 completed comparators and 70 disrupted comparators.
Top-line posture: Mixed precedent posture (completed fit 39.4%; disrupted fit 41.4%).
Most important design differential: Primary Purpose shows 50.6% completed-trial match versus 57.1% disrupted-trial match.
Lead action: No strong design outlier was detected in the current benchmark snapshot.
Interpretation note: this output is intended to support senior design review, not replace clinical, statistical, operational, or regulatory judgment.

15. Recommendation Detail

Full detail on all recommendations generated by this analysis run. Each recommendation includes a rationale derived from the precedent data and a specific evidence statement referencing the underlying statistics.

Priority	Category	Recommendation	Rationale	Evidence
Monitor	Overall review	No strong design outlier was detected in the current benchmark snapshot.	The reviewed draft does not currently show a clear signal that it is behaving materially worse than historical precedent.	Continue with clinical, statistical, operational, and regulatory review before final sign-off.

16. Comparator Exemplars

The table below lists representative studies from the matched cohort, separated by comparator lens (completed precedent, disrupted precedent, active context). These studies can be reviewed directly on ClinicalTrials.gov using the NCT ID provided.

Comparator Lens	Status	NCT ID	Title	Enrollment	Sponsor
Completed Precedent	Completed	NCT01012739	Efficacy, Safety, Tolerability, and Pharmacokinetics of Indacaterol Maleate Via Concept1 or Simoon Devices	35.0	Novartis Pharmaceuticals
Completed Precedent	Completed	NCT01959412	Crossover Study to Evaluate the Efficacy, Safety and Tolerability of Different Doses of Indacaterol in Patients With Persistent Asthma	91.0	Novartis Pharmaceuticals

Comparator Lens	Status	NCT ID	Title	Enrollment	Sponsor
Completed Precedent	Completed	NCT01820481	Safety and Efficacy of WIN-901X in Asthma	209.0	Whanin Pharmaceutical Company
Completed Precedent	Completed	NCT00980200	Efficacy and Safety Study in Subjects With Asthma	75.0	GlaxoSmithKline
Disrupted Precedent	Withdrawn	NCT01435902	A Study of a New Asthma Medicine in Asthmatics Whose Asthma Worsens With Exercise	0.0	GlaxoSmithKline
Disrupted Precedent	Withdrawn	NCT01923272	Non-Invasive Neurostimulation of the Vagus Nerve With the AlphaCore Device for the Relief of EIB	0.0	ElectroCore INC
Disrupted Precedent	Withdrawn	NCT01845623	A Study to Evaluate the Effects of Indacaterol Maleate (QAB149) as a New Formulation in the EPIC Test Fixture	0.0	Novartis Pharmaceuticals
Disrupted Precedent	Withdrawn	NCT03680976	Anti-Inflammatory Lipid Mediators in Asthma	0.0	Sally E. Wenzel MD
Active Context	Not Yet Recruiting	NCT06618105	Bronchoprovocation Study to Demonstrate the Pharmacodynamic Bioequivalence of Albuterol Sulfate HFA Inhalation Aerosol (E.Q. 90 mcg of Albuterol Base/Inh) of Macleods Pharmaceuticals Ltd	144.0	Macleods Pharmaceuticals Ltd
Active Context	Enrolling By Invitation	NCT06820749	A Study to Evaluate the Efficacy and Safety of HSK31858 Tablets in Patients with Airway Mucus Hypersecretion	309.0	Haisco Pharmaceutical Group Co., Ltd.

18. Audit Trail and Provenance

Every action taken in this session is recorded below with a UTC timestamp, actor, artifact type, and full detail string. This trail supports governance review, regulatory readiness, and reproducibility audits.

2026-03-19 17:53 UTC | Extract Protocol Profile

Actor: System | Artifact: Document | ID: Prot_002.pdf

Extracted protocol profile from Prot_002.pdf.

2026-03-19 17:54 UTC | Review Protocol Profile

Actor: User | Artifact: Protocol Profile

Reviewer confirmed and saved protocol profile fields.

2026-03-19 17:54 UTC | Build Design Similar Cohort

Actor: System | Artifact: Comparison Cohort | Metadata: condition: Asthma, condition_matched_total: 1000, design_similar_selected: 972, similarity_score_median: 0.345, sponsor_used_for_selection: False, completed: 626, disrupted: 70, evidence_strength: Strong, posture: Mixed precedent posture

Fetched 1000 condition-matched trials for 'Asthma'. Design similarity filter selected 972 trials (97.2% of pool). Dimensions used: population, design, endpoints, intervention, duration. Sponsor NOT used for selection.

2026-03-19 17:56 UTC | Export Pdf Report

Actor: System | Artifact: Report | ID: trial_protocol_report.pdf | Metadata: pubmed_articles: 0

Generated trial_protocol_report.pdf.

2026-03-19 17:56 UTC | Fetch Pubmed Evidence

Actor: System | Artifact: Literature Evidence | ID: Asthma | Metadata: article_count: 8, endpoint_focus: Other

Retrieved 8 PubMed articles for 'Asthma'.

2026-03-19 17:56 UTC | Export Pdf Report

Actor: System | Artifact: Report | ID: trial_protocol_report.pdf | Metadata: pubmed_articles: 8

Generated trial_protocol_report.pdf.

19. Methodology Notes

Source trials were retrieved from the ClinicalTrials.gov public registry (API v2) and normalised into a comparator cohort filtered to the protocol's clinical condition. Trials were classified as **completed** (status: COMPLETED) or **disrupted** (status: TERMINATED, SUSPENDED, or WITHDRAWN) and used as two distinct analytical lenses. Design alignment was assessed across seven domains: Phase, Study Type, Allocation, Masking, Intervention Model, Primary Purpose, and Endpoint Focus. Alignment scores represent the proportion of comparator trials sharing the protocol's design choice in each domain.

Protocol fields were extracted using deterministic heuristic parsing and, where configured, an LLM-assisted structured extraction step. PubMed evidence was retrieved via the NCBI E-utilities API (free access, no key required) using a condition-and-design-context query. All LLM calls are logged with model name and confidence metadata.

All outputs in this report are intended to support expert review and informed decision-making. They do not constitute a regulatory opinion, clinical recommendation, or approval decision. The report should be reviewed alongside clinical, statistical, operational, and regulatory expertise appropriate to the development programme.